Stereo-Lidar Fusion for Autonomous Driving

Computer Vision Course Project

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Presentation Workflow

Our discussion is structured into seven logical stages:

01-17 Foundations

➔**Context and Motivation:** Our team and competition.

➔**Problem Statement:** Racing requirements & sensor limits.

➔**State of the Art:** Monocular lifting vs. Grounded stereo.

➔**Proposed Method:** architecture principles.

➔**Dataset:** KITTI-360 & Small object transferability.

18-24 Results & Outlook

➔**Experimental Setup:** Hardware & Depth matcher study.

➔**Model Evaluation:** Single-sensor floors vs. Multimodal fusion (Quantitative & Qualitative).

➔**Conclusions:** Breakthroughs & FS Deployment.

fast charge

Context and Motivation

Fast Charge & Formula Student Driverless

Why are we developing advanced Stereo-LiDAR perception? **To enable high-speed autonomous racing** against the world's top engineering universities.

Formula Student Driverless

An elite international competition challenging universities to design, build, and race fully autonomous electric vehicles.

Sapienza Fast Charge

Our university's electric racing team. We are developing a custom 2WD electric prototype that relies heavily on its software stack to navigate unknown tracks.

The Mission (Why we do this)

The track has no lane lines, delimited entirely by color-coded traffic cones. Safe navigation requires **millimeter-accurate, far-range detection** of these small obstacles.

The Edge Compute Reality

The entire perception, SLAM, and control stack must run locally on a ruggedized **NVIDIA Jetson AGX Orin** under strict latency constraints while subjected to intense track vibrations.

Problem Statement

The Challenge: Why Multimodal 2D BEV?

Downstream Navigation Needs

Position (x,y) + Class
SLAM and Motion Planning require metric ground-plane coordinates.

Planar Bias
For Formula Student, height (z) and yaw are irrelevant for cone obstacles.

Deployment
Must operate in real-time on edge hardware (Jetson AGX Orin).

The Single-Sensor Dilemma

Camera (Stereo/RGB)
Pros: High semantic richness.
Cons: Quadratic depth error degradation at far range.

LiDAR
Pros: Precise metric geometry.
Cons: Sparse returns on small objects (In our case cones).

> **The Core Challenge:** Rescue sparse LiDAR returns with dense semantics while anchoring far-field depth with LiDAR geometry—efficiently.

State Of The Art

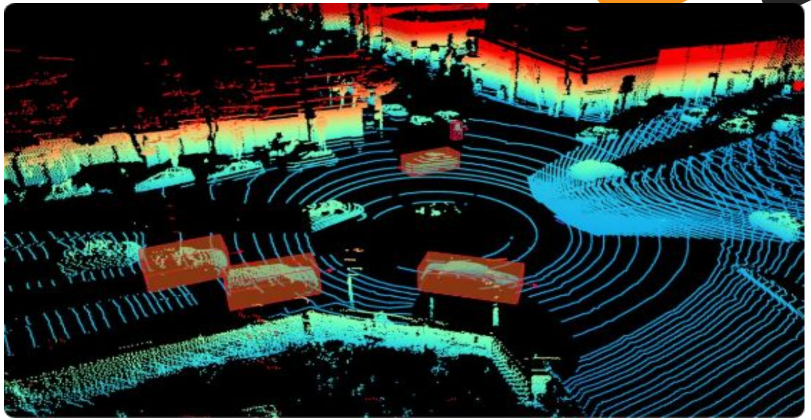
Monocular Bottlenecks in Mid-Fusion

1. Late Fusion (Box-Level)

Processes sensors independently. Cannot resolve blind spots and if LiDAR fails to trigger a box, camera semantics are ignored.

2. SOTA: Monocular "Lift-Splat-Shoot"

Fuses LiDAR with a monocular camera. Because monocular images lack native depth, the network must predict a per-pixel categorical depth distribution across bins to "lift" 2D image features into a 3D frustum before splatting them to BEV.



The monocular bottleneck:

Predicting depth from a single image is an ill-posed, error-prone guess. Errors in depth estimation smear image features across incorrect BEV cells, creating geometric "ghosts" that degrade multimodal alignment.

Our Innovation:

We replace predicted monocular depth with measured stereo depth. By leveraging geometric triangulation, feature lifting becomes a deterministic projection rather than a learned guess

Proposed Method

Governing Principles & Architecture

The Alignment Rule: Only fuse data in the same physical space. LiDAR and Stereo meet only in the canonical **Bird's-Eye-View (BEV) grid**.

The Unified 6-Stage Perception Pipeline

01. Camera Backbone
YOLO26 / EfficientNet features.

02. Splat to BEV
Grounded Stereo Frustum Projection.

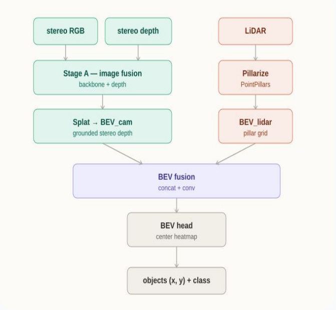
03. LiDAR Stem
Pillarize → PointPillars Scatter.

04. Mid-Fusion
Swappable (CNN vs. Transformer).

05. BEV Context
2D Spatial reasoning.

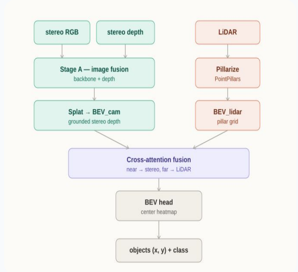
06. CenterPoint
Focal Heatmap + Offset Regressor.

BEV Grid Fusion



Direct voxel concatenation for preserving raw structural details.

Cross-Attention Fusion



Feature-level fusion using queries to focus on relevant spatial context.

Camera Branch: Grounded Stereo Projection

THE MONOCULAR BOTTLENECK (`MonoBEV`)

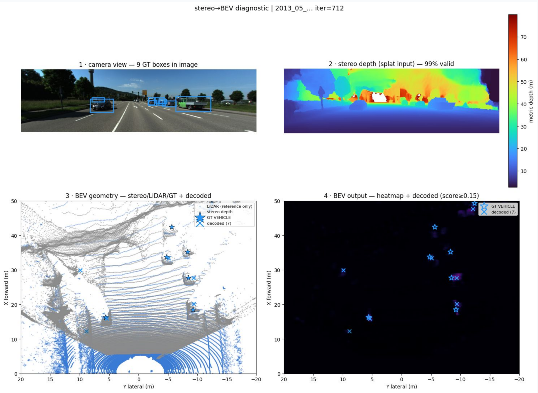
Standard Lift-Splat-Shoot predicts soft probability over depth bins. This creates a massive frustum outer-product cloud, incurring heavy memory overhead and ghosting when depth guessing fails.

OUR SOLUTION: GROUNDED METRIC LIFTING

Replaces guessed distributions with **hard metric depth** from stereo matching. Every valid feature pixel is deterministically unprojected into 3D ego-coordinates.

DETERMINISTIC VOXEL SPLATTING

Uses a fast sort and cumulative-sum algorithm (`splat`) to aggregate points into the BEV grid, outputting a clean **64 ch** feature map.



KEY ADVANTAGE

Eliminating the **d-bin outer product** drastically reduces GPU memory consumption and guarantees that visual semantics land in their exact physical BEV cells!

Lidar Branch

Why PointPillars for Racing?

• **Computational Efficiency:** Standard 3D voxelization requires slow 3D convolutional networks.

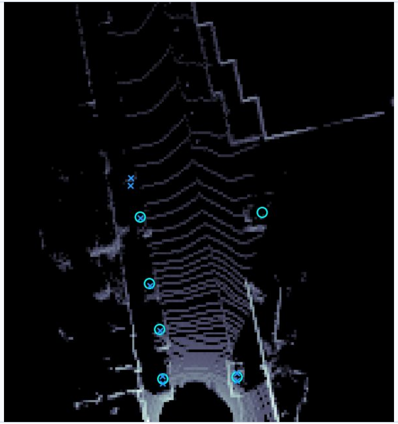
• **Real-Time Execution:** We adopt PointPillars to organize point clouds into vertical columns of infinite height, eliminating Z-axis binning on Jetson AGX Orin.

The 3-Step Pillar Pipeline

01. Pillarization (`pillarize`): Points are grouped into a 2D grid (up to 32 points per pillar, 12,000 max non-empty pillars).

02. Point Decoration: Raw returns are augmented to 9D vectors by encoding geometric offsets to the arithmetic mean and center.

03. Feature Net & Scatter: Simplified PointNet generates a 64-ch embedding, scattered to 2D coordinates to output 128 ch.



KEY HIGHLIGHT

Zero 3D Convolutions: By collapsing the vertical dimension into 2D pseudo-images, our LiDAR stem operates entirely via fast 2D convolutions, achieving sub-millisecond encoding!

Pipelines A: ConcatConvFusion

Pipeline A (Primary)

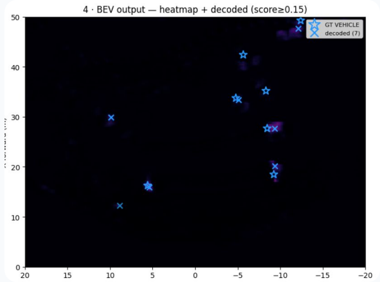
- **Channel Concatenation**
  Camera (64ch) & LiDAR (128ch) stacked (192ch).
- **Preservation**
  Allows mixing without forced scale addition.
- **Block**
  2-layer 3×3 Conv stack with BatchNorm/ReLU.

CONSTRAINTS

Relies on strict calibration alignment. Sensitive to physical sensor drift during racing.

Optimal for static, rigid racing rigs with perfect calibration.

OUTPUT



BEV Output Heatmap

Camera & LiDAR stacked (192ch) detection output with score ≥ 0.15.

Pipeline C: CrossAttentionFusion

Windowed Cross-Attention

→ **Relaxed Alignment:** Camera (Query) searches over LiDAR (Keys/Values).

→ **Compute:** Local 8×8 windows reduce complexity by **500x** vs. global attention.

→ **Equivariance:** Learnable Relative Position Bias.

Near/Far Spatial Gating

Sigmoid gate modulates trust dynamically:

$$F_{fused} = (1 - g) \cdot F_{cam} + g \cdot F_{attn}$$

Trusts dense stereo in **Near Field (≤20m)** and shifts to attended LiDAR geometry in **Far Field (>50m)**.

Deep Dive: Pipeline A vs Pipeline C

Metric	ConcatConv (A)	Cross-Attn (C)
Complexity	Linear O(N)	Local Windowed
Parameters	~369 k	~450 k
Tolerance	Strict Alignment	Relaxed Search

Engineering Trade-Off: Pipeline A is optimal for rigid rigs; Pipeline C for rough terrain/vibrations.

Dataset

KITTI-360 & Small Object Regime

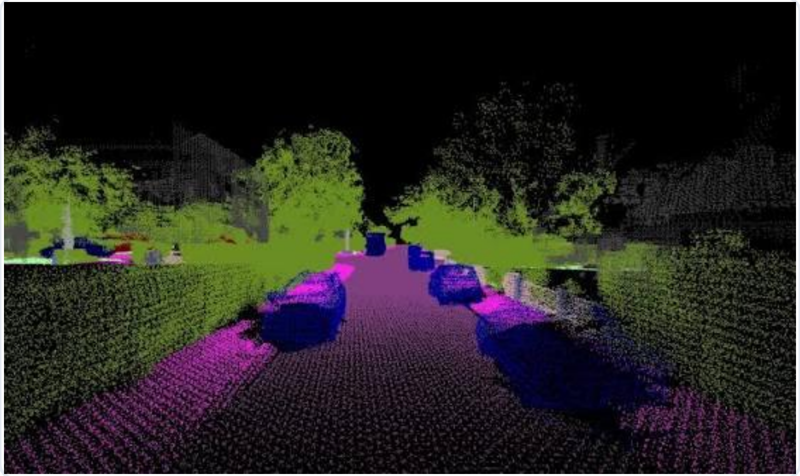
Why KITTI-360?

The only unified dataset providing:

→ True **Color Stereo Pairs**.

→ Dense **64-beam LiDAR** (~114k pts).

→ Accurate **3D Bounding Boxes**.



Why Small Classes (Sign, Person, Bike)? Exact geometric proxy for Formula Student Cones where camera semantics rescue detection.

Experimental Setup

Training Configuration

Hardware

RTX 5050 Mobile
RTX 4070 Mobile
RTX 2060

Loss Function

Focal Heatmap +
Offset L1

Optimizer

AdamW

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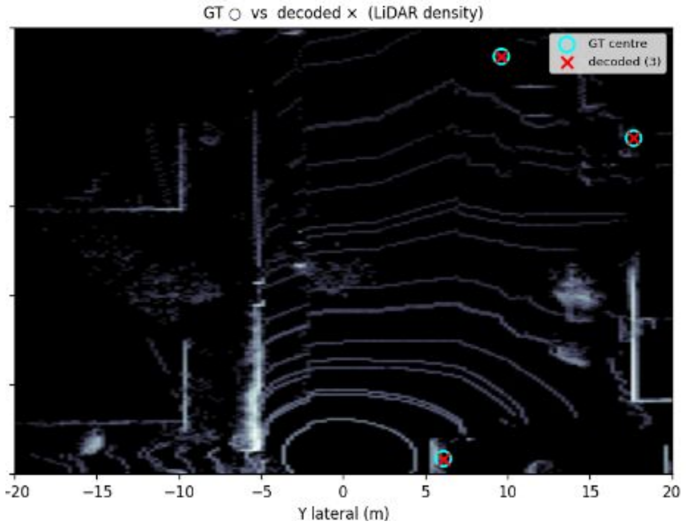
Model Evalutation

Quantitative Comparison

#	Method	Params	mAP ↑	VEH AP@2	SIGN AP@2	P ↑	R ↑	F1 ↑	Err (m) ↓
0	LiDAR-only (full drives, 5-class) †	223k	0.222	0.588	0.269	0.314	0.274	0.286	**0.468**
1	Camera baseline (SGBM, p3, frozen)	772k	0.184	0.492	0.147	0.279	0.274	0.273	0.648
2	+ IGEV stereo (p3)	772k	0.190	0.521	0.163	0.284	0.295	0.284	0.683
3	EfficientNet-scratch (nms fix)	1.07M	0.142	0.420	0.130	0.284	0.224	0.250	0.707
4	YOLO p3p4	805k	0.224	0.603	0.212	0.368	0.299	0.328	0.614
5	YOLO p3 + WD 1e-4	805k	0.225	0.573	0.166	0.362	0.301	0.327	0.617
6	YOLO p3p4p5	870k	0.215	0.656	0.158	0.318	0.337	0.323	0.648
7	MonoBEV (predicted depth)	1.37M	0.066	0.252	0.000	0.146	0.116	0.129	1.000
8	Depth-ctx + p3p4	809k	0.214	0.644	0.185	0.367	0.316	0.338	0.673
9	Depth-ctx + p3p4 + WD + Dropout	809k	0.227	0.631	0.196	**0.371**	**0.390**	**0.351**	0.613
10	Pipeline A (fused, concat)	1.36M	0.265	0.684	0.217	0.359	0.333	0.339	0.473
11	Pipeline C (fused, cross-attn)	1.42M	**0.276**	**0.701**	**0.232**	0.344	0.375	0.350	0.476

Qualitative Comparison



Complementary Coverage

Camera populates near-field; LiDAR anchors far-field geometry.

False Positive Suppression

Mid-fusion clears visual shadows that trigger camera-only alarms.

Sub-Cell Precision

Predicted centers (red) align tightly within GT boxes (green).

Conclusions & Future Work

Methodology & Racing Roadmap

Conclusions

Grounded lifting eliminates monocular error. Localization AP@0.5m improved by **+15%**. Windowed attention enables real-time edge transformer fusion even during vibrations.

FS Porting Roadmap

01 Swap IGEV for **ZED X hardware depth**.

02 Re-tune for **16-beam VLP-16 LiDAR**.

03 Fine-tune on our **Cone Dataset**.

04 Deploy on **Jetson AGX Orin**.

References

Scientific Literature

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Liu et al., "Multi-Task Multi-Sensor Fusion", ICRA 2023.

02 **IGEV-Stereo**
Xu et al., "Iterative Geometry Encoding Volume", CVPR 2023.

03 **SLBEVFusion**
"3D detection using stereo camera and LiDAR", Neurocomputing, 2024.

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Liao et al., "Novel Dataset and Benchmarks", TPAMI 2022.

05 **FutrTrack**
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06 **PointPillars**
Lang et al., "Fast Encoders for Object Detection", CVPR 2019.

07 **UniCorrn**
Goswami et al., "Unified Correspondence Transformer Across 2D and 3D", CVPR 2026.

08 **Lift, Splat, Shoot**
Philion et al., "Encoding Images from Arbitrary Camera Rigs by Implicitly Unprojecting to 3D.", ECCV 2020.

Thanks for your attention!